ISCB-LATAM SolBio CCBCOL 2024

Spodoptera frugiperda virome (corn strain) based on RNA-seq: identification of potential bio-controls

Natalia López Rivera[1], Karol Landines[2], Carolina Montoya-Ruiz[2], Clara Saldamando[1]

[1]Universidad Nacional de Colombia, Grupo de Biotecnología Vegetal. Facultad de Ciencias, Medellín, Colombia.
[2]Universidad Nacional de Colombia, Grupo de Biotecnología Vegetal. Facultad de Ciencias de la Vida, Medellín, Colombia.

BACKGROUND

Spodoptera frugiperda is a pest that causes devastating damage primarily to corn and rice crops in Colombia, resulting in millions of dollars in economic losses worldwide. Due to the wide distribution of the corn strain and its increasing resistance to insecticides, the search for more sustainable control strategies is urgent. Among these, the use of microbiota and insect-associated viruses presents a promising alternative. Viruses, especially baculoviruses, are common in lepidopterans and widely used in biotechnology due to their genetic manipulability and stability (Páez et al., 2020). They are excellent candidates for pest biocontrol because of their ability to suppress insect populations long-term without multiple interventions and their high species specificity (Hussain et al., 2021). However, in Colombia, limited knowledge about *S. frugiperda*-associated viruses restricts the development of innovative management strategies for this pest

Figure 1. *Spodoptera frugiperda* betabaculovirus

OBJECTIVE

Characterize the virome of larvae *Spodoptera frugiperda* corn strain to identify viruses with biocontrol potential

METHODOLOGY

Three larvae of S. frugiperda (instar 5) were collected in Envigado, Antioquia. Their RNA was extracted using the TRIzol method. RNA quality was verified and sequenced (pair-end) with the Illumina NovaSeq X plus platform at CD Genomics (New York)

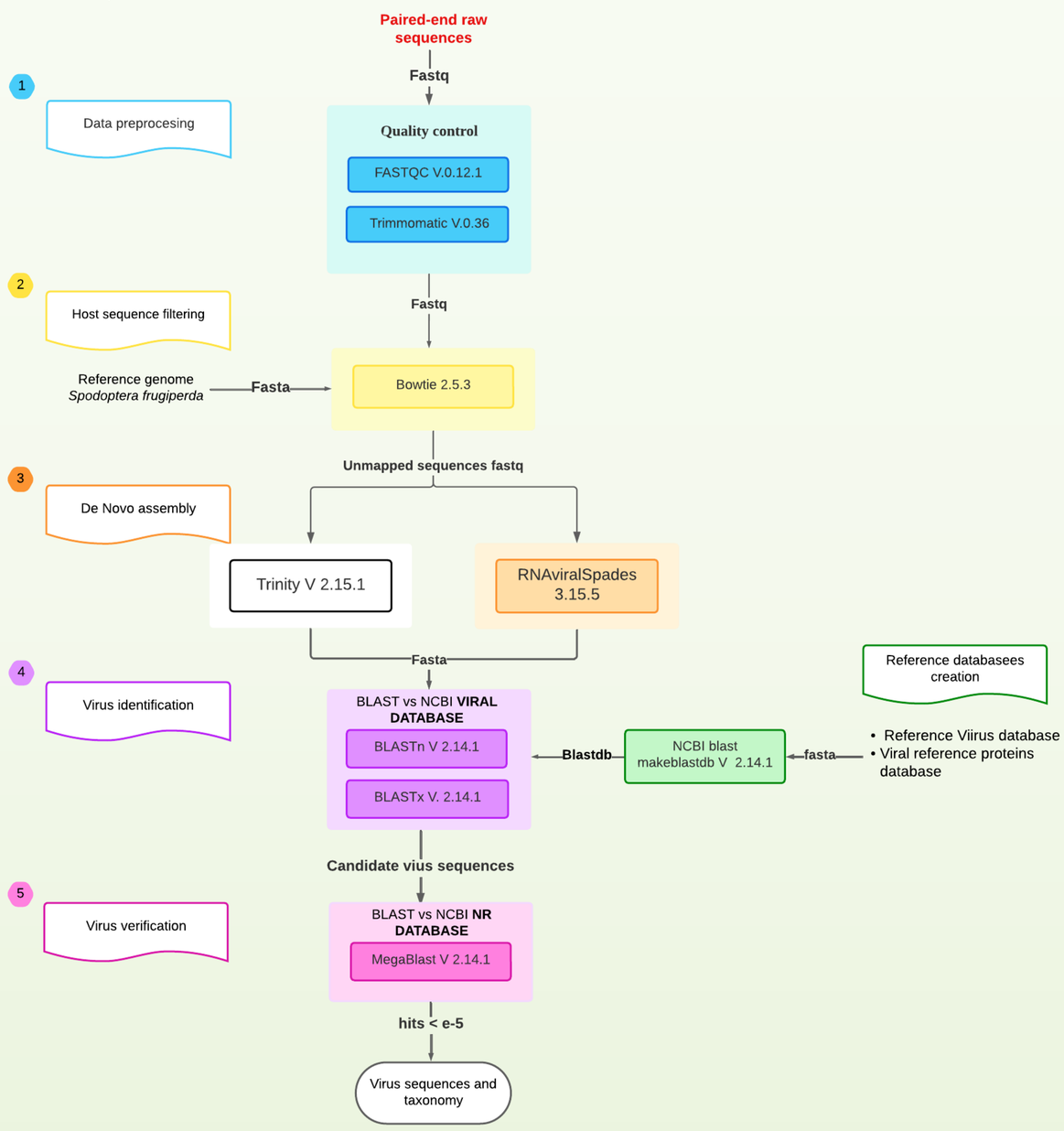

Figure 2. Bioinformatic pipeline; Each processes are shown in a specific color (blue for preprocessing, yellow for filtering, orange for assembly, purple for virus identification and pink for virus verification) and the software used are displayed in rectangles

PRELIMINAR RESULTS AND DISCUSSION

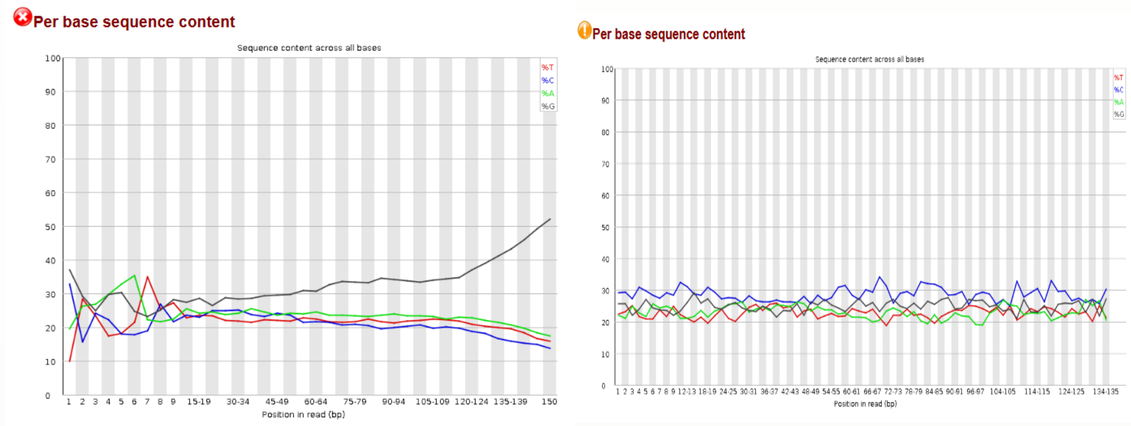

Figure 3. Quality Analysis. Low-quality nucleotides that could interfere with the analyses were removed.

Table 1. Difference in assembly programs. More transcripts were identified with the rnaviralspades program than with Trinity, as it is specifically designed for viruses.

	Trinity	RNAviralSpades
Dataset1	58.650	109.953
Dataset2	77.813	95.405
Dataset3	28.521	38.871

De novo assemblies were performed with Trinity and rnaviralSpades to compare approaches: **Trinity** optimizes complete transcriptome reconstruction, while **rnaviralSPAdes**, being designed for viral RNA, generated more transcripts in virus-rich samples

Table 2. BLASTn results. It was performed with an e-value of $1x10^{-5}$ against a reference virus database downloaded from NCBI.

Query accession	Identity percentag	Alignment length	# gap openings	E-value	Virus information	Query Coverage Per Subject
NODE_13341_length_472_	70,8	192	0	5,87E-18	NC_026511,1 |Spodoptera frugiperda Betabaculovirus isolate VG008, complete genom	63
NODE_13341_length_472_	72,0	107	0	4,23E-07	NC_026511,1 |Spodoptera frugiperda Betabaculovirus isolate VG008, complete genom	63
NODE_24439_length_355_	73,4	192	0	1,32E-24	NC_026511,1 |Spodoptera frugiperda Betabaculovirus isolate VG008, complete genom	54
NODE_31015_length_320_	72,0	211	2	6,13E-22	NC_026511,1 |Spodoptera frugiperda Betabaculovirus isolate VG008, complete genom	66
NODE_35486_length_302_	72,4	196	0	4,72E-23	NC_026511,1 |Spodoptera frugiperda Betabaculovirus isolate VG008, complete genom	65
NODE_36055_length_300_	69,0	229	2	1,87E-15	NC_026511,1 |Spodoptera frugiperda Betabaculovirus isolate VG008, complete genom	76
NODE_45439_length_271_	71,5	186	2	1,37E-16	NC_026511,1 |Spodoptera frugiperda Betabaculovirus isolate VG008, complete genom	68
NODE_52678_length_254_	70,7	188	0	3,64E-17	NC_026511,1 |Spodoptera frugiperda Betabaculovirus isolate VG008, complete genom	74
NODE_2994_length_993_c	73,4	233	2	1,46E-29	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	67
NODE_2994_length_993_c	73,5	204	0	7,55E-27	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	67
NODE_2994_length_993_c	70,6	231	2	2,47E-20	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	67
NODE_13341_length_472_	70,0	230	0	1,13E-20	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	84
NODE_13341_length_472_	70,6	170	0	3,71E-14	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	84
NODE_24439_length_355_	68,8	231	0	1,51E-17	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	65
NODE_31015_length_320_	69,5	197	0	2,00E-15	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	62
NODE_35486_length_302_	70,1	197	0	1,54E-16	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	65
NODE_36055_length_300_	70,7	215	0	6,95E-21	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	91
NODE_52678_length_254_	69,7	188	0	1,89E-14	NC_009011,2 |Spodoptera frugiperda MNPV virus, complete genome	74

Initially, viral sequences from the "*Betabaculovirus*" family and "*multi-nucleopolyhedrovirus* (MNPV)" were found. Most were filtered based on identity percentages above 70% and alignment lengths over 100 nucleotides.

Table 3. Inconsistencies in Viral Sequence Verification. The MEGABLAST verification against the NCBI non-redundant database did not retrieve the previously identified viruses.

Virus information vs VIRAL DATABASE	Virus information vs NR DATABASE
NC_011616,1 |Spodoptera litura nucleopolyhedrovirus II, complete genome	PREDICTED: Spodoptera frugiperda Kv channel-interacting protein 1 (LOC118278646), transcript variant X1, mRNA
NC_011616,1 |Spodoptera litura nucleopolyhedrovirus II, complete genome	PREDICTED: Spodoptera frugiperda uncharacterized LOC118279803 (LOC118279803), mRNA
NC_011616,1 |Spodoptera litura nucleopolyhedrovirus II, complete genome	PREDICTED: Spodoptera frugiperda MAP kinase-activated protein kinase 2 (LOC118280217), transcript variant X1, mRNA
NC_011616,1 |Spodoptera litura nucleopolyhedrovirus II, complete genome	Spodoptera frugiperda SINE SfSE1.57, complete sequence
NC_011616,1 |Spodoptera litura nucleopolyhedrovirus II, complete genome	Spodoptera frugiperda SINE SfSE1.15, complete sequence

CONCLUSIONS

1. It is necessary to evaluate additional transcriptome analysis programs for future analyses.
2. Determining the cause of the inconsistencies in the results is essential to ensure more reliable outcomes and facilitate subsequent analyses

REFERENCES

Páez, David J., and Arietta E. Fleming-Davies. 2020. Understanding the Evolutionary Ecology of Host Pathogen Interactions Provides Insights into the Outcomes of Insect Pest Biocontrol. Viruses 12 (2). https://doi.org/10.3390/V12020141.

Hussain, Ahmed G., Jörg T. Wennmann, Georg Goergen, Astrid Bryon, and Vera I.D. Ros. 2021. ¿Viruses of the Fall Armyworm Spodoptera Frugiperda: A Review with Prospects for Biological Control.¿ Viruses 13 (11): 1¿21. https://doi.org/10.3390/v13112220.

Zhao, G., Wu, G., Lim, E. S., Droit, L., Krishnamurthy, S., Barouch, D. H., Virgin, H. W., & Wang, D. (2017). VirusSeeker, a computational pipeline for virus discovery and virome composition analysis. Virology, 503, 21-30. https://doi.org/10.1016/j.virol.2017.01.005